H1 Geography Prelim Suggested Answers

Cluster 2: Fieldwork

1 (a) Suggest how the students' hypothesis can be improved. [3]

Award 1 mark for each suggestion on how the hypothesis can be improved, to a maximum of 3 marks.
Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Explanation [1m] and Further Development [+1m]
• A **suitable spatial scope** i.e. (Bukit Panjang) would improve the hypothesis as the students had collected data in one neighbourhood in Singapore. **[1m]**
• **Precise age-group definitions** would improve the hypothesis because "youths" and "other age groups" are vague, so exact numerical age ranges should be stated to create clear, mutually exclusive **independent-variable categories**. **[1m]** • Furthermore, **precise age-group definitions** would enable all six students to classify participants consistently across the three return points, reducing **observer variation** and improving the **reliability** of comparisons between age groups. **[+1m]**

(b) With reference to Resources 1A and 1B, explain **one strength** and **one weakness** of the selected return points. [4]

Award 1 mark for each explanation on the strength or weakness, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation of strength or weakness, where applicable.
Award maximum of 2 marks for either strengths or weaknesses only.

Explanation [1m] and Further Development [+1m]
• The **land-use diversity** of the selected return points is a strength because Resources 1A and 1B show sites at *a hawker centre and market*, *a FairPrice supermarket outlet* and *an HDB void deck*, allowing participation to be studied across contrasting everyday settings. **[1m]** • For example, *Return Point X at Bukit Panjang Hawker Centre and Market* may capture diners and market visitors, while *Return Point Y at FairPrice* captures shoppers, possibly **increasing the range of age groups** included in the observations. **[+1m]**
• The selected return points are located in areas with **high pedestrian footfall**, such as *Bukit Panjang Hawker Centre and Market* and *FairPrice supermarket*, increasing the likelihood of observing a **larger number of potential participants** within the data-collection period. **[1m]** • This can increase the **sample size and reliability of the observations**, as more people are exposed to the return points, reducing the likelihood that findings are based on only a small number of users. **[+1m]**

(c) Suggest how students could minimise the impact of their investigation at the three return points. [4]

Award 1 mark for each suggestion on how students could minimise the impact of their investigation, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Explanation [1m] and Further Development [+1m]
• **Avoiding obstruction to pedestrian movement** would minimise the investigation's impact because students should stand beside, rather than directly in front of, return machines, entrances and walkways, preventing **congestion** and allowing users to move safely. **[1m]** • For example, at **Return Point Y**, students should approach respondents only after they have returned their containers and move away from the supermarket shelves to avoid inconveniencing customers or disrupting business operations. **[+1m]**
• **Minimising disturbance to the public** would reduce the investigation's social impact because students should conduct observations quietly and administer questionnaires only to willing participants without interrupting shopping, dining or other daily activities. **[1m]**
• **Leaving the sites clean** would minimise the investigation's **environmental impact** because students should collect all worksheets, stationery and personal waste after data collection instead of leaving litter that could reduce the cleanliness of public spaces. **[1m]**

(d) Describe how the data shown in Resource 2 may be represented. [4]

Award 1 mark for each description on how the data may be represented to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Graph type	Description [1m] and Further Development [+1m]
Compound bar graph	• A **compound bar graph** could place *Return Points X, Y and Z* on the horizontal axis, while the vertical axis shows the **number of people** using a scale from zero to at least 36. **[1m]** • Each compound bar should represent the *total number of people* recorded at its return point, with bar heights of *36 at X, 28 at Y and 24 at Z*. **[+1m]** • Each bar should be subdivided into **youth** and **non-youth** sections, using values of *18 and 18 at X, 7 and 21 at Y, and 10 and 14 at Z*. **[+1m]** • Contrasting shading and a clear **legend**, title and axis labels should distinguish youths from non-youths, allowing both the total number and youth composition at each return point to be compared. **[+1m]**

(e) Explain how the questionnaire shown in Resource 3 may be improved. [5]

Award 1 mark for each explanation on how the questionnaire may be improved, to a maximum of 5 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Explanation [1m] and Further Development [+1m]
• The **age-group classification** in Resource 3 should be made **exhaustive** and **mutually exclusive** because *"0–7 years", "10–16 years", "20–25 years" and "Above 25 years"* omit people aged 8–9 and 17–19, preventing complete and accurate **data collection**. **[1m]** • For example, the questionnaire could use continuous categories such as 0–9, 10–19, 20–29 and 30 years or above, ensuring every respondent can be classified consistently for **comparison**. **[+1m]**
• The **respondent coverage** should be widened because the questionnaires were administered only to youths, however the hypothesis compares youths' participation with that of other age groups, producing an unrepresentative **sample** for testing the stated relationship. **[1m]** • For example, at all three Bukit Panjang return points, students should administer the same questionnaire to youths and adults from each identified age group, allowing direct comparison of their reported **participation rates**. **[+1m]**
• The **question validity** of Resource 3 should be improved because Q1 asks *"Which age group do you think is more likely to use the Return Points?"*, measuring respondents' perceptions rather than their actual **recycling behaviour**. **[1m]** • For example, respondents at the Bukit Panjang return points could instead be asked how many times they personally returned beverage containers during the previous month, generating measurable evidence directly linked to **participation**. **[+1m]**

(f) Evaluate the data collection methods for this investigation. [10]

Possible approaches:

Candidates could evaluate the investigation by reviewing the methods used to collect data and the data presented in Resources 1A, 1B, 2 and 3.

Candidates could review the counting procedure at each return point and/or assess all three sites collectively. Evaluation could include whether collecting data concurrently from 08:00 to 11:00 on one Friday morning improved comparability, whether the three student groups applied consistent age-group classifications, and whether differences between observers affected the accuracy and reliability of the counts.

Candidates could evaluate the site selection and scale of the investigation using Resources 1A and 1B. This may include considering whether the three return points represent the range of return-point environments in Bukit Panjang, whether differences in accessibility and surrounding land use affected participation, and whether findings from one neighbourhood can represent participation across Singapore.

Candidates could also evaluate the sampling method used for the questionnaire survey. Surveying youths who were present and willing to participate is efficient, but it is a voluntary or convenience sample that may exclude less willing youths and those who did not visit the return points during the investigation period. Candidates could also consider whether surveying only youths provides sufficient evidence for a hypothesis that compares youths with other age groups.

For Resource 3, candidates could consider the clarity, neutrality and relevance of the questionnaire items. They may assess whether the questions measure actual participation in the Beverage Container Return Scheme or only respondents' stated attitudes and intentions, and how any unclear wording or limited response options could affect the validity of the conclusions drawn.

For the data in Resource 2, candidates could evaluate whether the observed numbers and proportions provide sufficient evidence to compare age groups. They could consider the sample size at each return point, whether percentages should be used where site totals differ, and whether observations should be repeated across different days and times.

Levels marked using **Generic Level Descriptors for H1 fieldwork evaluative questions**.

Section B

Cluster 1: Sustainable Future and Climate Change

Answer **both** questions.

2 Resource 4A shows Warton Road in Stratford, East London, in 2005, before reimaging. Resource 4B shows the same location in 2012, after reimaging. Resource 5 shows the Olympic Park and its surrounding areas in East London during the London 2012 Olympic Games and three years after the Games, in 2015. Resource 6 shows changes in selected land uses in East London, from 2005 to 2015.

(a) Explain the economic reasons for urban reimaging efforts. [4]

Award 1 mark for each explanation of the economic reasons for urban reimaging efforts, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Explanation [1m] and Further Development [+1m]
• **Addressing the economic consequences of deindustrialisation** is a reason for **urban reimaging efforts** because authorities can replace **declining manufacturing industries** with new **service-sector activities**, thereby restoring employment, business activity and economic growth. **[1m]**
• **Attracting tourism revenue** is a reason for **urban reimaging efforts** because a distinctive and appealing city image attracts more visitors, whose expenditure on accommodation, dining, entertainment and attractions generates income for the **local economy**. **[1m]**
• **Attracting investment** is a reason for **urban reimaging efforts** because modern infrastructure, attractive urban environments and improved amenities increase a city's competitiveness for **foreign direct investment**, supporting economic development and employment. **[1m]** • For example, Jurong Lake District was reimaged as a modern **mixed-use district** with waterfront attractions, green spaces and enhanced infrastructure, helping the area attract over SGD 10 billion in investment. **[+1m]**

(b) Using Resources 4A and 4B, describe the changes to the built environment along Warton Road, Stratford from 2005 to 2012. [3]

Award 1 mark for each description on the changes to the built environment, to a maximum of 3 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Explanation [1m] and Further Development [+1m]
• **Building quality and architectural design** have changed as Resource 4A shows *a plain, weathered brick industrial wall*, whereas Resource 4B shows *a modern multi-storey building with colourful panels and a redesigned external facade*. **[1m]**
• **Street cleanliness and physical condition** improved because Resource 4A shows *two discarded refrigeration units and scattered debris beside the road*, while Resource 4B shows *a cleared, cleaner and more orderly street environment*. **[1m]**
• **Access and security infrastructure** became more developed because Resource 4A shows *an open road without a controlled entrance*, whereas Resource 4B shows *a security booth, barrier arms, bollards and traffic signs*. **[1m]**

(c) Explain how the changes to the built environment as shown in Resources 4A and 4B may improve the liveability of Stratford, East London. [5]

Award 1 mark for each explanation of how the changes to the built environment as shown in Resources 4A and 4B may improve the liveability, to a maximum of 5 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Explanation [1m] + Further Development [+1m]
• The **economic** aspects of **liveability** may improve because Resource 4B shows *modern, visually distinctive buildings and upgraded access infrastructure*, replacing the *older industrial surroundings* visible in Resource 4A and making Stratford more attractive to businesses and investors. **[1m]** • Furthermore, the *modernised buildings and better-maintained streetscape in Resource 4B* could encourage new commercial activities, creating **employment opportunities** and increasing residents' access to goods and services. **[+1m]**
• The **environmental** aspects of **liveability** may improve because Resource 4A shows *two discarded refrigeration units and roadside litter*, whereas Resource 4B shows *a cleaner, and more orderly streetscape without bulky waste*. **[1m]** • In addition, the **environmental** aspect of **liveability** may improve because removing bulky waste reduces **exposure to** possible physical/biological hazards, creating a more hygienic and safe **urban environment** for Stratford's residents. **[+1m]**
• The **social** dimension of **liveability** may improve because Resource 4B shows *a security booth, vehicle barriers, bollards and directional signs provide greater road safety for* Stratford's residents by regulating vehicle movement. **[1m]**

(d) With reference to Resources 5 and 6, explain how the London 2012 Olympic Games has contributed to sustainable urban development. [6]

Award 1 mark for each explanation of how the London 2012 Olympic Games has contributed to sustainable urban development, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Explanation [1m] and Further Development [+1m]
• The **economic** dimension to **sustainable urban development** is shown in Resource 5, where *business and commercial land increased from approximately 20 hectares in 2005 to 50 hectares in 2015*, creating space for sustained commercial activity and employment. **[1m]** • Furthermore, the improvement in the **economic** dimension is evident beyond the Games because Resource 6 shows that *the former broadcast and media centres were converted into Here East (businesses)*, ensuring the long-term productive reuse of Olympic infrastructure. **[+1m]**
• The **environmental** dimension to **sustainable urban development** is evident in Resource 5, where *derelict industrial land decreased from approximately 160 hectares in 2005 to about five hectares in 2015*, substantially reducing urban dereliction. **[1m]** • Furthermore, **environmental** sustainability was strengthened because Resource 5 shows that *parks and open space increased from approximately 25 hectares to around 100 hectares*, creating more **green urban spaces**. **[+1m]**
• The **social** dimension to **sustainable urban development** is shown in Resource 5, where *housing land increased from approximately 10 hectares in 2005 to 55 hectares in 2015*, expanding residential space for East London's population. **[1m]** • In addition, Resource 6 shows that *the Olympic Village became East Village, alongside new housing at Chobham Manor and East Wick*, providing lasting residential uses beyond the Olympics. **[+1m]**

(e) Using named example(s), suggest why urban reimaging strategies may be unsuccessful. [4]

Award 1 mark for each suggestion on why urban reimaging strategies may be unsuccessful, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Explanation [1m] and Further Development [+1m]
• The **Rio de Janeiro 2016 Olympic Park** illustrates weak **economic sustainability** because inadequate **post-event planning** left expensive Olympic venues underused, reducing their ability to generate lasting revenue, employment and returns on investment. **[1m]** • Furthermore, the **Rio de Janeiro 2016 Olympic Park** experienced theft and vandalism at venues such as the Maracana Stadium and Olympic Aquatics Stadium, causing costly infrastructure to deteriorate after the Games. **[+1m]**
• **Palm Jumeirah** illustrates weak **environmental sustainability** because constructing the artificial archipelago required extensive **dredging** and **land reclamation**, destroying marine habitats and imposing substantial ecological costs on surrounding coastal ecosystems. **[1m]** • Furthermore, **Palm Jumeirah** altered natural water circulation during construction, disrupting marine ecological processes and reducing the project's contribution to long-term **environmental sustainability** despite its enhanced global image. **[+1m]**

3 Resource 7 shows global sea surface temperature anomalies, March 2024. Resource 8 shows an article on the impacts of climate change on aquatic ecosystems. Resource 9 shows projected global mean sea-level rise under different greenhouse gas emissions scenarios.

(a) Using Resource 7, describe the global sea surface temperature anomalies in March 2024. [4]
Award 1 mark for each description of the global sea surface temperature anomalies, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable

Explanation [1m] and Further Development [+1m]
• Most of the world's oceans experienced **positive sea surface temperature anomalies**, with temperatures above the 1991–2020 average, as indicated by the widespread yellow, orange and red shading. [1m]
• The highest positive anomalies (approximately **+2°C to more than +3°C**) were concentrated in parts of the **equatorial Atlantic Ocean, eastern tropical Pacific Ocean, and parts of the Indian Ocean.** [1m]
• Negative sea surface temperature anomalies (approximately **−0.5°C to −2°C**) occurred in limited areas, mainly in parts of the **South Pacific Ocean, Southern Ocean**, and around parts of the **Antarctic region and South Atlantic Ocean**. [1m]
• Although positive anomalies dominated globally, a few isolated regions still recorded below-average sea surface temperatures, indicating that warming was not spatially uniform. **[1m]**

(b) Suggest how contemporary climate change may accelerate warming through positive feedbacks. [3]

Award 1 mark for each suggestion on how contemporary climate change may accelerate warming through positive feedbacks, to a maximum of 3 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Explanation [1m] and Further Development [+1m]
• Rising temperatures cause glaciers and ice sheets to melt, reducing the Earth's albedo. More solar radiation is absorbed by darker land or ocean surfaces, leading to further warming and additional ice melt. [1] • In the Arctic, declining sea ice exposes darker ocean water, increasing solar absorption and Arctic warming, which accelerates further sea-ice melt. [+1m]
• Higher temperatures thaw permafrost, releasing greenhouse gases such as methane and carbon dioxide into the atmosphere. These gases enhance the greenhouse effect, causing further warming. [1]

(c) With reference to Resource 8, explain how changes in aquatic ecosystems as a result of contemporary climate change can affect humans. [7]

Award 1 mark for each explanation of how changes in aquatic ecosystems can affect humans, to a maximum of 7 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Explanation [1m] and Further Development [+1m]
• Resource 8 states that coral bleaching and the degradation of coral reefs reduce habitats and breeding grounds for many fish species, resulting in lower fish populations and reduced fish catches. [1m] • Lower fish catches reduce the productivity of fisheries, affecting the livelihoods and incomes of fishing communities that depend on fishing as their primary source of employment. [+1m]
• Resource 8 explains that ocean acidification reduces the survival of shell-forming organisms and plankton, disrupting marine food chains and reducing biodiversity. [1m] • As plankton and other marine organisms form the base of many food chains, declines in their populations reduce the abundance of commercially important fish species. [+1m]
• Resource 8 states that deteriorating coral reefs become less effective at reducing wave energy, making coastlines more exposed to wave action and storm surges. [1m] • Greater exposure to wave action increases coastal erosion and the risk of flooding, threatening homes, transport networks and other coastal infrastructure. [+1m]
• Resource 8 shows that coral bleaching and reef degradation reduce the visual attractiveness and biodiversity of coral reefs, decreasing opportunities for recreational activities such as diving and snorkelling. [1m] • Reduced tourist interest lowers tourism revenue and employment for coastal communities dependent on reef-based recreation, while also diminishing the cultural and aesthetic value of marine environments. [+1m]

(d) Using Resource 9, compare possible pathways for future sea level rise under different greenhouse gas emissions scenarios. [4]

Award 1 mark for each comparison of possible pathways for future sea level rise under different greenhouse gas emissions scenarios, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable. Award maximum of 2 marks if either similarities or differences only.

Explanation [1m] and Further Development [+1m]
• All greenhouse gas emissions scenarios project that global sea level will continue to rise from 2020 to 2100, with none showing a decline in sea level. **[1m]** • All projected pathways show that sea level rises gradually at first before increasing more rapidly towards the end of the century. **[1m]** • None of the scenarios shows sea level stabilising before 2100. **[1m]**
• Higher greenhouse gas emissions scenarios project much greater sea-level rise than lower emissions scenarios. By 2100, the high emissions scenario reaches about 6.7 feet, whereas the low emissions scenario reaches only about 1 foot. **[1m]**

(e) Suggest reasons why projections of future sea-level rise remain uncertain. [4]

Award 1 mark for each explanation of why projections of future sea-level rise remain uncertain, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Explanation [1m] and Further Development [+1m]
• Future greenhouse gas emissions depend on uncertain factors such as population growth, economic development, technological advances and the effectiveness of climate mitigation policies. [1m] • Different greenhouse gas emissions pathways result in different rates of global warming, leading to varying rates of thermal expansion of seawater and melting of glaciers and ice sheets. Consequently, projections of future sea-level rise remain uncertain. [+1m]
• Scientists do not fully understand complex physical processes, such as the behaviour of the Greenland and Antarctic ice sheets, glacier melting and climate feedback mechanisms. [1m] • Since these processes influence the rate and magnitude of sea-level rise, uncertainties in how they respond to future warming reduce the accuracy of sea-level rise projections. [+1m]

Section C

Cluster 1: Sustainable Future and Climate Change

4 Evaluate the impact of multiple deprivations experienced by slum dwellers on sustainable urban development. [13]

Possible Approaches:

Candidates could approach the question by making a judgement on whether multiple deprivations have a greater impact on sustainable urban development in some places than others through a consideration of criteria such as the spatial scale of the impacts, the temporal scale of the impacts and the aspects of sustainable urban development affected. Candidates could analyse the contextual factors such as the scale and location of slums, effectiveness of urban governance and resources available, which may have contributed to the variations in the impacts.

Candidates could also approach the question by making a judgement on whether some forms of deprivation have a greater impact on sustainable urban development than others through a consideration of criteria such as the number of people affected, the severity of the deprivation and the ability of slum dwellers to overcome its negative impacts. Candidates could consider different forms of deprivation related to economic well-being, social well-being and environmental well-being. Candidates could also analyse how different forms of deprivation interact and reinforce one another.

Levels marked using **Generic Level Descriptors for H1 essays**

5 'There is greater complementarity than tension between sustainable urban development and liveability.'

Evaluate this statement. [13]

Possible Approaches:

Candidates could approach the question by making a judgement on whether the complementarity between sustainable urban development and liveability is greater than the tension through a consideration of criteria such as the scale of the benefits and costs, the aspects of liveability affected and the stakeholders impacted. Candidates could consider strategies related to transport, housing, green infrastructure, waste management and the provision of public spaces. Candidates could also consider strategies implemented at different scales or by different stakeholders.

Candidates could also approach the question by making a judgement on whether complementarity is greater than tension in some contexts than others through a consideration of two or more case studies where both complementarity and tension are present. Candidates could analyse the contextual factors such as the level of economic development, quality of urban governance, availability of land and distribution of costs and benefits, which may have contributed to differences in the relative significance of complementarity and tension. Candidates could also consider whether the relationship varies between different groups of urban residents or over different temporal scales.

Levels marked using **Generic Level Descriptors for H1 essays**

6 Evaluate the success of adaptation strategies in responding to climate change. [13]

Possible Approaches:

Candidates could approach the question by making a judgement on whether some adaptation strategies are more effective than others through a consideration of the relative strengths and limitations of the strategies. The strategies could be compared in relation to criteria such as costs, benefits and ability to reduce exposure and vulnerability to climate change. These strategies could be aimed at responding to different climate-related impacts such as flooding, sea-level rise, drought and extreme temperatures. They could also include those implemented at different scales or by different stakeholders.

Candidates could also approach the question by making a judgement on whether the same adaptation strategies have been more effective in some places than others through a consideration of two or more case studies. Candidates could evaluate whether the strategies in each case achieved the intended aims. Candidates could also analyse the unintended negative impacts of the strategies. Candidates could also analyse the contextual factors such as the magnitude of climatic hazards, political will of the authorities and resources available, which may have contributed to the variations in success between these places.

Levels marked using **Generic Level Descriptors for H1 essays**